



Service Tool Instruction

Service Tool Instruction
Number: 3377897-2

Revision Level:

Released Date: 13-feb-
2003

Description

Cylinder Head Leak Test Kit

Purpose

This document provides instructions for using the cylinder head leak test kit, Part Number 3164341. This kit can be used with QST30, K19, K38, K50, QSK19, QSK19G, QSK23, QSK45, QSK45G, QSK60, QSK60G, and QSK78 Series engines.

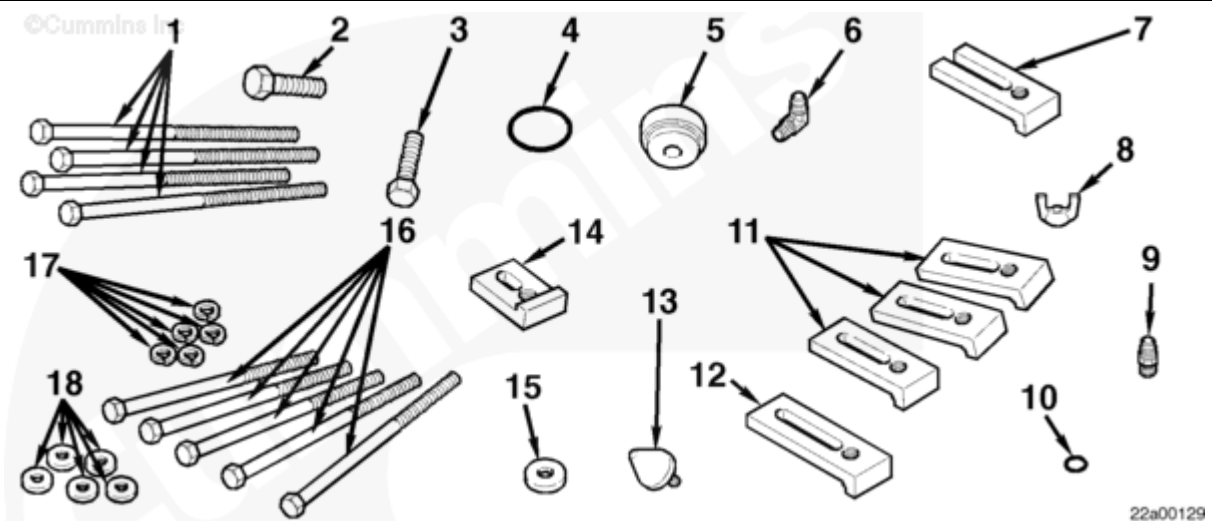


Table 1, Cylinder Head Leak Test Kit, Part Number 3164341

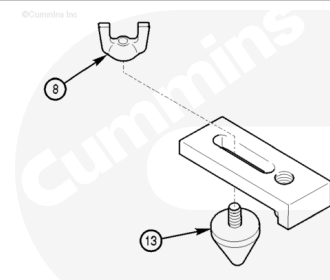
Item Number	Part Number	Description	Quantity
1	S141C	Hex head capscrew, 3/8-16 x 6.75 inch	4
2	S145	Hex head capscrew, 1/2-13 x 1.25 inch	1
3	3908545	Hex head capscrew, M12 x 1.75 x 40-mm	1
4	3164372	O-ring, 1-3/16 x 1-3/8 (1 shown)	2
5	3164340	Test adapter	1

Table 1, Cylinder Head Leak Test Kit, Part Number 3164341			
Item Number	Part Number	Description	Quantity
6	S-1047-A	Adapter elbow, #4 x 1/8 inch x 45-degree male	1
7	3164339	Clamp	1
8	S295	Wing nut, 1/4-20 (1 shown)	5
9	3164370	Pipe nipple, #4 x 10 x 1.25-mm	1
10	3963982	Sealing washer (1 shown)	2
11	3164337	Clamp, 3.0 inch	3
12	3164338	Clamp, 3.75 inch	1
13	3164371	Plug (1 shown)	10
14	3164336	Clamp, 1.5 inch	1
15	S-670	Washer, 1/2 inch	1
16	69621	Hex head capscrew, 3/8-16 x 5.5 inch	5
17	S-604	Lock washer, 3/8 inch	5
18	3003163	Washer, 3/8 inch (extra thick)	5

Table 2, Items Used with Cylinder Head Leak Test Kit, Purchased Separately			
Item Number	Part Number	Description	Quantity
Not shown	3164231	Pressure regulator valve kit	1

Preparatory

Assemble the wing nut (8), clamp, and plug (13), as necessary.



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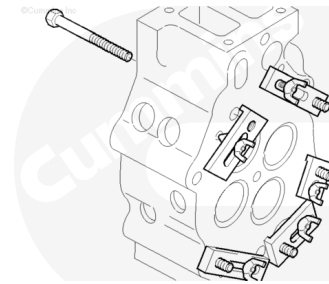
Note : The QST30 cylinder head as shown uses five clamp and plug assemblies; other models use only the four longer assemblies.

Install the clamp and plug assembly into the cylinder head, blocking the coolant passages.

Insert the appropriate capscrew through the valve guide side of the cylinder head, and thread it into the clamp.

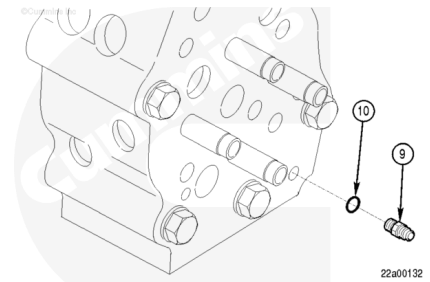
Tighten the capscrew.

Torque Value: 20 n•m [177 in-lb]



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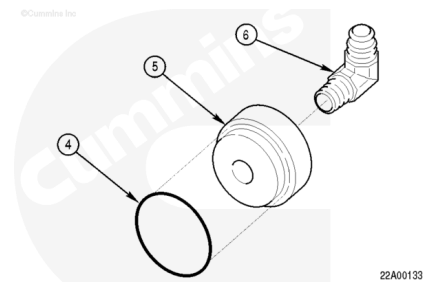
Install the sealing washer (10) and pipe nipple (9) into the coolant vent port of the cylinder head.



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Install the o-ring (4) into the o-ring groove of the test adapter (5).

Thread the adapter elbow (6) into adapter (5).

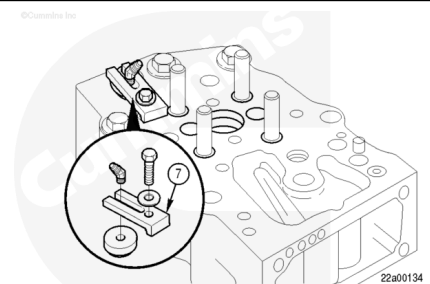


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Place the test adapter assembly into the cylinder head.

Install the clamp (7) over the test adapter assembly and secure it with the appropriate capscrew and washer.

Torque Value: 54 n•m [40 ft-lb]

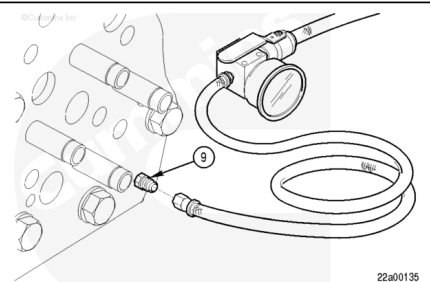


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⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Connect the pressure regulator valve kit to the pipe nipple (9).



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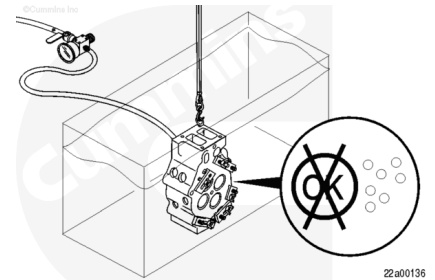
Connect the pressure regulator valve kit to shop air and adjust air pressure.

Measurements

	kpa	psi
Air Pressure:	345	50

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.



Note : The QST30 Series engine cylinder head is shown.

Attach an eyebolt to the cylinder head.

Use a suitable lifting device to lift the cylinder head.

Immerse the cylinder head into a tank of water.

Check for bubbles. Bubbles indicate an air leak.

If the cylinder head leaks, it **must** be repaired or replaced.